

Polyamide 6/carbon black/molybdenum disulphide composites: Friction, wear and morphological characteristics

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HIGHLIGHTS

- Introduction of MoS₂ with carbon black has reduced the wear of polyamide 6.
- The specific wear rate of polyamide 6/CB/MoS₂ composites decreased.
- The coefficient of friction of polyamide 6 decreased with increase in MoS₂ content.
- The tensile properties of polyamide 6 increased linearly with increase in MoS₂ content.

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ABSTRACT

The objective of this research work is to evaluate the tribological and physico–mechanical properties of molybdenum disulphide (MoS₂) loaded polyamide 6 composites in the presence of carbon black (CB). Polyamide 6 containing 1 wt% of carbon black powder was compounded and extruded with varying amounts viz., 0.5, 1.0, 2.0 and 3.0 wt%, MoS₂ in a co-rotating twin screw extruder. The fabricated polyamide 6 composites have been evaluated for tribological properties such as wear loss, specific wear rate and coefficient of friction, and physico–mechanical properties such as water uptake, density, surface hardness, tensile properties and impact strength. The effect of molybdenum disulphide content, abrading loads, sliding velocities and sliding distances on wear characteristics of the polyamide 6/CB/MoS₂ composites were evaluated using a pin-on-disc equipment. It was found that the introduction of MoS₂ has significantly reduced the wear loss and friction behavior of pristine polyamide 6. Worn surfaces were examined by scanning electron microscopy (SEM) to have better insight on the wear mechanism. Furthermore the polyamide 6 composites showed significant improvement in tensile behavior. Additionally, the thermal characteristics of the composites also have been studied by using differential scanning calorimetry (DSC), thermogravimetric analysis (TGA) and dynamic mechanical analysis (DMA).

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1. Introduction

Polyamides (nylons), the most favored class of engineering materials, especially polyamide 6 and polyamide 66, are preferred as materials for bearing, gears, cams etc., by virtue of their easy availability, appreciably good combination of strength along with tribo-performance and cost. Polyamide's toughness, low coefficient of friction and good abrasion resistance make it an ideal replacement for a wide variety of materials from metal to rubber. Using polyamide reduces lubrication requirements, eliminates galling,

corrosion and pilferage – problems, and improves wear resistance and sound dampening characteristics. It was reported that the friction and wear behavior of polyamides was fairly satisfactory under dry sliding conditions and lubrication at higher speeds. However, in order to keep pace with the modern technological innovations, ever increasing demands are being placed on tribo-materials for enhanced performance for operating under stringent conditions of abrading loads, speeds, temperatures and hazardous environments. In order to enhance the tribological characteristics of polyamides efficiently, solid lubricants have been added into the polymer matrix. A solid lubricant is defined as a material that provides lubrication, under essentially dry conditions, to two surfaces moving relatively to each other. The solid lubricants often lead to decrease of friction coefficient and wear

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rate through the reduction in adhesion with the counterface or the creation of a transfer film with low shear strength at the interface [1,2]. There are many research reports on the improvement in wear resistance of various polyamides such as polyamide 6, polyamide 66, polyamide 11, polyamide 46, etc., with solid lubricants such as molybdenum sulphide (MoS_2), graphite (Gr) and polytetrafluoroethylene (PTFE) [3,4]. These are the predominant materials used as solid lubricants in thermoplastics.

MoS_2 is a black crystalline sulfide of molybdenum and it has a layered structure. In its appearance and feel, MoS_2 is similar to graphite. It is a dark blue-gray or black solid, which feels slippery or greasy to the touch. It has hexagonal layer lattice. It is widely used as a solid lubricant because of its low friction properties, sometimes at relatively high temperatures. MoS_2 with particle sizes in the range of 1–100 μm is a common dry lubricant. MoS_2 is often a component of blends and composites where low friction is sought. When added to plastics, MoS_2 forms a composite with improved strength as well as reduced friction. MoS_2 and graphite have a layered structure. The important effect is that the materials can shear more easily parallel to the layers than across them. They can therefore support relatively heavy loads at right angle to the layers while still being able to slide easily parallel to the layers. This property is being effectively used for lubrication process. The coefficient of friction is more or less equal to the shear stress parallel to the layers divided by the yield stress or hardness perpendicular to the layers. Because the low friction only occurs parallel to the layers, it follows that these solid lubricants will only be effective when their layers are parallel to the direction of sliding. It is also important that the solid lubricant should adhere strongly to the bearing surface; otherwise it would be easily rubbed away and gives very short service life. Yinping et al. [5] have reported that high load and frequency promote the formation of a compact transfer film. The compact transfer films are believed to be the predominant mechanism giving rise to high load-carrying capacity and excellent wear-resistance of the bonded MoS_2 solid film lubricants [5,6].

The importance of the tribological properties of polymer composites and blends convinced various researchers to study the wear behavior and to improve the wear resistance of polymeric materials. Due to the low coefficient of friction and also the ability to maintain loads, some specific grades of polymers are used in place of the traditional metal based materials in recent times [7,8]. There are reports in the literature [8,9] on the slide wear of several polymers sliding against a steel counter surface showed that the wear loss increased with increasing load/speed and wear rate decreased with sliding distance. The decrease in wear rate is caused by progressive smoothening of the surface and by the formation of a protective transfer film of polymer on the steel counter surface [10,11]. The load and sliding speed showed a stronger effect on the wear rate of the composites [3]. Wear performance of materials depends on the inherent material properties such as physical and chemical of the ingredients and on the operating conditions such as pressure, speed, temperature, environment, counterface etc., [12].

Reinforcements preferentially enhance the wear resistance while solid lubricants reduce the friction. However, efforts to optimize the combination of these fillers to boost the strength and tribo-performance of polyamide 6 in different wear modes are not systematically investigated in detail and hence, required especially in the background of such papers available for other polymers and composites [13–16]. In this research investigation, with an objective to derive the benefits, molybdenum disulphides (MoS_2) powder as well as carbon black (CB), loaded polyamide 6 composites were fabricated by varying the concentrations of the MoS_2 from 0.5 to 3 wt% with CB constant at 1 wt%. To improve the tribological performance, normally the polymer matrix should possess a high temperature resistance or it should dissipate the

heat to withstand the high heat generated during the friction/abrasion. Additions of fillers that absorb and dissipate the heat are often of great advantage, especially if effects of temperature enhancement in the contact area must be avoided in order to prevent an increase in the specific wear rate. Carbon black is one such conductive filler which forms conductive pathways in the polymer matrix to dissipate heat which is generated due to frictional sliding. CB is a form of amorphous carbon that has high surface area to volume ratio. It is being used as pigments and filler. But it is not much explored in the tribological performances of thermoplastics like polyamides. The fabricated polyamide 6/CB/ MoS_2 composites have been evaluated for tribological properties such as wear and friction and physico-mechanical properties. The thermal characteristics of the composites have been studied by using differential scanning calorimetry (DSC), thermogravimetric analysis (TGA) and dynamic mechanical analysis (DMA). The effect of MoS_2 content, abrading loads, sliding velocities and sliding distances on wear characteristics of the polyamide composites were evaluated using a pin-on-disc equipment.

2. Experimental

2.1. Materials

A laboratory grade polyamide 6 (DuPont Zytel, 7300T) and molybdenum disulphide (MoS_2) powder (Starplex) were used. Carbon black powder obtained from Graphite India with particle sizes in the range 50–60 μm was used as received. Polyamide 6 was dried at 80 °C for 8 h prior to use to remove the possible moisture.

2.2. Compounding and specimen preparation

Carbon black powder (1 wt%) was premixed with varying amounts viz., 0.5, 1, 2 and 3 wt% of MoS_2 powder and then mixed with polyamide 6 granules in tumbling mixer for 15 min, after pre-drying in hot air oven at 80 °C for 8 h duration, and then melt blended using a co-rotating intermesh twin screw extruder (Nanjing Rubber & Plastics Pvt. Ltd., China) at a screw speed of 175 rpm with barrel temperature ranging from 250 to 260 °C. The extruder consists of nine nozzles and the temperature zones maintained at each of the nozzles are different and lay in the range 270–280 °C. The extrudate strand was pelletized and stored in sealed packs containing desiccant. The test specimens for tensile behaviors, impact strength, and water absorption were prepared using an R.H. WINSOR INDIA, SD-75 automatic injection molding machine with 70 tons clamping pressure at 260–285 °C and an injection pressure of 80 bars. After molding, the test specimens were conditioned at 23 ± 2 °C and $50 \pm 5\%$ RH for 40 h according to ASTM D 618 prior to testing.

2.3. Testing and characterization techniques

The tribological properties of polyamide 6/CB/ MoS_2 composites were carried out by using a pin-on-disc wear testing machine, model Ducom LR20E of India, developed according to ASTM G99-04 method [17–19]. The counterface disc was made of stainless steel (AISI 314). The disc of 150 mm diameter, thickness of 8 mm, surface roughness of 25 μm and hardness of 62 HRC was used for evaluating the sliding wear properties of the composites. The test specimens were weighed and initial weights were recorded using a high precision digital electronic balance after thorough cleaning. After recording initial weight, the specimen was fixed to the holder such that the flat face of the specimen comes in contact with the rotating hardened steel disc. The setup had an arrangement to vary the motor speed and consequently the rpm of the disc. The sliding

Table 1
Coefficient of friction for polyamide 6/MoS₂/CB composites at velocity of 5 m s⁻¹.

Load (N)	Sliding length (m)	Coefficient of friction for polyamide 6/MoS ₂ /CB composites with varying MoS ₂ (%)				
		0	0.5	1.0	2.0	3.0
50	1000	0.165	0.158	0.152	0.141	0.130
	1500	0.195	0.186	0.181	0.175	0.168
	2000	0.214	0.202	0.196	0.183	0.179
100	1000	0.181	0.175	0.164	0.153	0.142
	1500	0.227	0.213	0.204	0.192	0.185
	2000	0.239	0.228	0.219	0.206	0.193
150	1000	0.205	0.189	0.176	0.168	0.155
	1500	0.243	0.232	0.226	0.213	0.208
	2000	0.255	0.242	0.231	0.222	0.216

wear characteristics were performed for varying amounts of MoS₂ content at different sliding velocities of 5, 7 and 9 m s⁻¹; at different sliding distances of 1000, 1500 and 2000 m; and at varying applied loads of 50, 100 and 150 N. The final weights of the specimens were recorded and the wear loss, specific wear rate and coefficient of friction were investigated. Generally the specific wear rate is defined as the wear volume normalized by the normal load and the sliding distance. The specific wear rate (K_S , g N⁻¹ m), was calculated from the equation;

$$K_S = W/F_N \times d \quad (1)$$

where, W is the weight loss in grams, F_N is the normal load in Newton; and d is the sliding distance in meters. The worn surfaces were cleaned with acetone and thoroughly dried, and then the worn surfaces were observed using a scanning electron microscope (SEM, JSM-5600LV, Japanese Jeol Co.), to elucidate the wear mechanisms, all the worn surfaces were plated with gold coating prior to SEM observation for rendering electric conductivity.

Water uptake measurements were made in 50 mm-diameter disc specimens as per ASTM D 570. Density measurements were made using Mettler Toledo electronic balance (model AG 204, Switzerland) according to ASTM D 792. Hand operated durometer was used to measure the surface hardness (Shore D) of the specimens according to ASTM D785. The tensile behavior of the composites was measured using JJ Lloyds Universal Testing Machine (model Z20, 20 KN, USA) as per ASTM D-638 method at a crosshead speed of 50 mm min⁻¹. Izod impact strength (notched) was measured in a WinPEN CEAST S.P.A., Italy according to ASTM D 256B method.

The transition temperatures of the samples were examined by using DSC (model DSC-Q 200, DuPont TA instrument, USA). All samples were sealed in hermetic aluminum pans. The dynamic DSC

scans were recorded at the heating rate of 10 °C min⁻¹ from ambient to 300 °C under nitrogen gas flow of 60 ml min⁻¹. The TGA thermograms were obtained using a DuPont TA instrument, TGA 2950 (USA) thermal analyzer at a heating rate of 20 °C min⁻¹ in nitrogen atmosphere. The TGA profiles were recorded over a temperature range of 30–700 °C. The weight of the samples used for each analysis was 6–8 mg. The storage modulus of polyamide 6/CB/MoS₂ composites has been studied by DMA. The DMA was carried out using NETZSCH DMA 242 instruments in 3 point bending mode from 25 °C to 200 °C. The testing frequency was 1 Hz and the heating rate was 5 °C min⁻¹ in nitrogen atmosphere.

3. Results and discussion

3.1. Coefficient of friction

The variation in the coefficient of friction as a function of MoS₂ content at 5 m s⁻¹ sliding velocity for MoS₂ filled polyamide 6/carbon black composites is tabulated in Table 1. The 'coefficient of friction' (COF), μ , is a dimensionless scalar value which describes the ratio of the force of friction between two bodies and the force pressing them together. The coefficient of friction depends on the materials used. A common way to reduce friction is by using a lubricant. One of the mechanisms of the corresponding reduction in the coefficient of friction is the formation of a transfer film on the surface of the counterpart [10]. It is also important that the solid lubricant should adhere strongly to the bearing surface; otherwise it would be easily rubbed away and leads to very short service life. Yinping et al. [5] have reported that high load and frequency promote the formation of a compact transfer film. The compact transfer films are believed to be the predominant mechanism giving rise to high load-carrying capacity and excellent

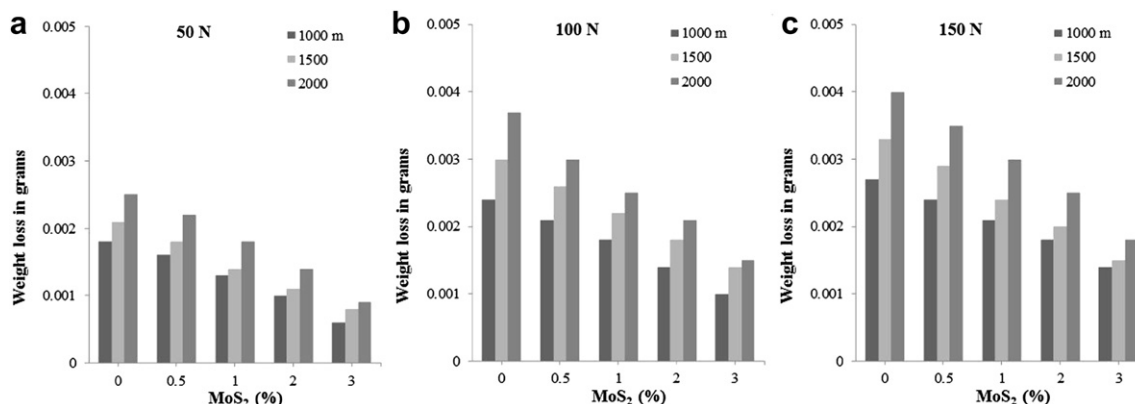


Fig. 1. Weight loss as a function of MoS₂ content for polyamide 6/CB/MoS₂ composites at varying sliding distances and at varying loads; (a) 50 N, (b) 100 N and (c) 150 N.

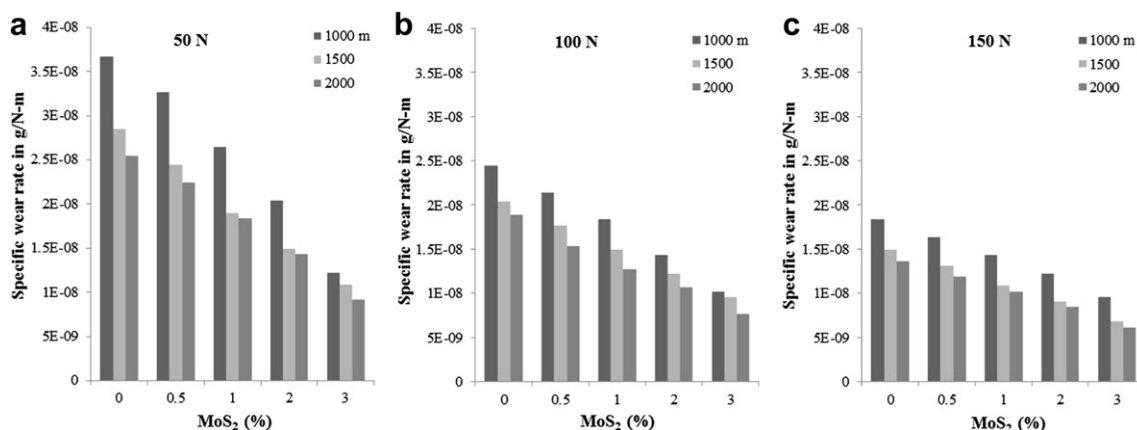


Fig. 2. Specific wear rate as a function of MoS₂ content for polyamide 6/CB/MoS₂ composites at varying sliding distances and varying loads; (a) 50 N, (b) 100 N and (c) 150 N.

wear-resistance performances of the bonded MoS₂ solid film lubricants [5,6]. The coefficient of friction decreases with increase in MoS₂ content (Table 1). During dry-sliding, the metallic counterface disc comes in contact with the polymer composite surface containing MoS₂ filler. Due to friction, the MoS₂ material comes out as fine powder and forms a thin film between the metallic counterface and the polymeric surface. This MoS₂ film was sticking to the metallic counterface rather than polymer composite surface. This thin film is called transfer film, which prevents further damage on the abrading (polymer) composite surface. This behavior is obviously relevant to the lubricating effect of MoS₂ which can reduce the adhesion between the polyamide composite and the metallic counterpart. MoS₂ is composed of sheets and layers. The layers themselves are strong but the bonding between the layers is weak. Consequently MoS₂ is strong in compression but weak in shear. This is advantageous for producing low friction [20–25]. As the real area of contact and shear strength of polymer substrate changes during sliding, the coefficient of friction increases with increase in sliding load. Similar trends were observed at other sliding distances and velocities investigated during the current studies.

3.2. Sliding wear behavior and specific wear rate

The plots of wear loss as a function of MoS₂ content (0, 0.5, 1, 2, and 3 wt%) at different applied loads (50, 100 and 150 N) and sliding distances (1000, 1500 and 2000 m) for all polyamide 6/CB/MoS₂ composites are shown in Fig. 1(a)–(c). All the plots indicate that wear loss decreases with increase in MoS₂ content. Specific wear rate as a function of MoS₂ content (0, 0.5, 1, 2, and 3 wt%) at

different applied loads (50, 100 and 150 N) and sliding distances (1000, 1500 and 2000 m) for all polyamide 6/CB/MoS₂ composites are shown in Fig. 2(a)–(c). In all these plots, specific wear rate is inversely proportional to MoS₂ content, applied load and sliding distance. It is obvious that the wear loss of composites decreases with increasing MoS₂ content but increases with increasing applied load and sliding distances. The improvement in wear resistance is due to the presence of solid lubricant (MoS₂) dispersed in the polymer matrix. These MoS₂ dispersed in the polymer matrix acts as barrier and also prevent large scale fragmentation of the polyamide 6/CB matrix. This behavior is clearly observed from SEM images. The MoS₂ also acts as reinforcing element, bears the load and reduces the wear rate. The wear resistance is maximum for 3 wt% MoS₂ loaded polyamide composite. The wear loss of MoS₂ filled polyamide 6/CB composites exhibiting an inverse relationship with all the parameters as the MoS₂ content goes on increases in the composites.

3.3. Surface morphology

Scanning electron microscopy (SEM) images were used for correlating the wear loss and wear rate data. SEM images of worn out surfaces of polyamide 6/CB without MoS₂, subjected to wear at 5 m s^{−1} sliding velocity for sliding distance of 1000 m at three different loads such as 50, 100 and 150 N, are shown in Fig. 3(a)–(c) respectively. It shows severe wear track and surface damage. However, the incorporation of MoS₂ reduces the wear and surface damage as seen in figures Fig. 4(a)–(d). Fig. 4 shows that the surface smoothness increases with increase in MoS₂ dosage. The worn surface of 0.5 wt% MoS₂ filled polyamide 6/CB composites is

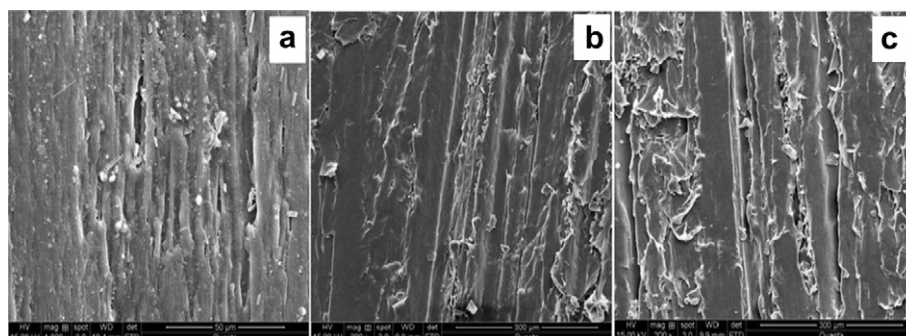


Fig. 3. SEM photomicrographs of worn out surfaces of polyamide 6/CB/MoS₂ composites at 5 m s^{−1} sliding velocity for 1000 m sliding distances at three loads (a) 50 N, (b) 100 N and (c) 150 N.

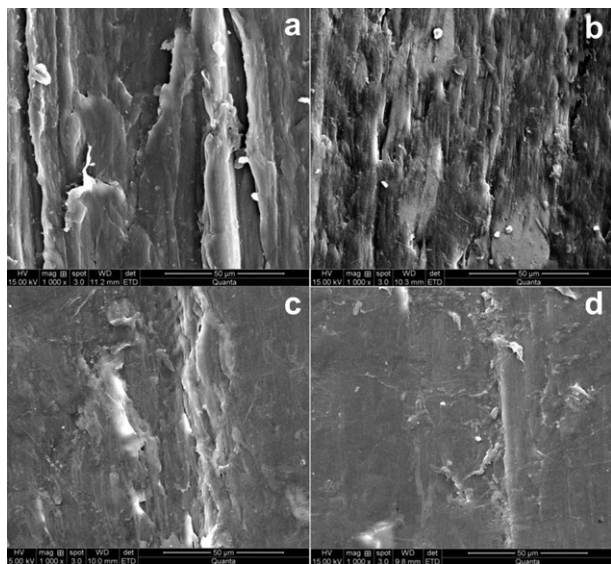


Fig. 4. SEM images of polyamide 6/CB/MoS₂ composites at 5 m s^{−1} sliding velocity, at 100 N load for 1000 m sliding distance and with varying amounts of MoS₂ content; (a) 0.5%, (b) 1.0%, (c) 2% and (d) 3%.

relatively rough with more matrix damage compared to other loadings (1.0, 2.0 and 3.0 wtMoS₂). There is evidence of more matrix removal and deep furrows (Fig. 5(a)–(b)) in the direction of sliding due to the plowing action of sharp counterface. The worn surface of 3 wt% of MoS₂ filled polyamide 6/CB composites (Fig. 5(d)) becomes smooth and smooth indicating that the wear volume loss is less, because of more and more MoS₂ particles adhered to the surface of the specimens hinders the abrasion of the matrix. It can be concluded from the SEM photomicrographs that inclusion of MoS₂ as a solid lubricant in the polyamide 6/CB matrix is beneficial from the sliding wear resistance point of view. Fig. 6(a)–(b) shows the sliding wear surfaces of MoS₂ filled polyamide 6/CB composites at a sliding velocity of 9 m s^{−1} for 1000 and 2000 m sliding distances at 150 N load. Fig. 7(a) shows the damaged matrix at higher sliding velocity and sliding distance indicating more material removal during abrasion process. As the sliding

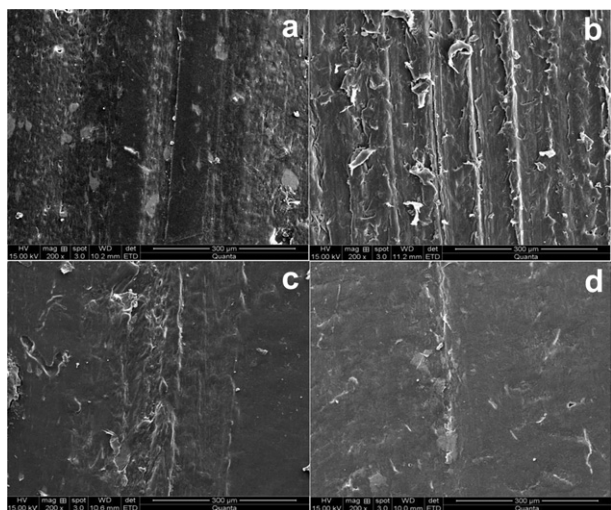


Fig. 5. SEM images of polyamide 6/CB/MoS₂ composites at 5 m s^{−1} sliding velocity, at 50 N load for 1000 m, sliding distance and with varying amounts of MoS₂ content; (a) 0.5%, (b) 1.0%, (c) 2% and (d) 3%.

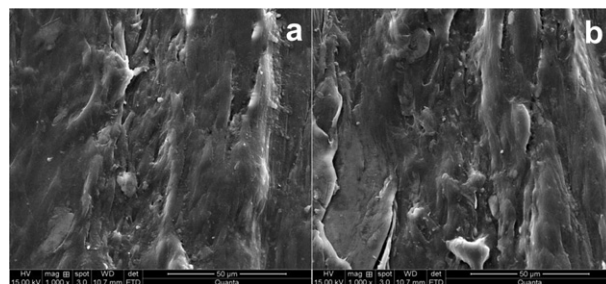


Fig. 6. SEM images of 0.5% MoS₂ filled polyamide 6/CB composite at 9 m s^{−1} sliding velocity, at 150 N load (a) 1000 m and (b) 2000 m.

distance increases from 1000 to 2000 m, the worn surface (Fig. 7(b)) becomes rough and which indicates that the wear volume loss is more. The worn surface images of higher sliding distance (2000 m) specimen exhibited the severe matrix damage.

3.4. Physico–mechanical properties

The fabricated polyamide 6/CB/MoS₂ composites have been characterized for physico–mechanical properties according to ASTM methods. The measured physico–mechanical properties such as water uptake, density, surface hardness, tensile behaviors and impact strength for polyamide 6/CB/MoS₂ are given in Table 2. The results shown in Table 2 indicate that the incorporation of MoS₂ has reduced the water uptake from 2.7 (0.5% MoS₂) to 2.2% (3% MoS₂). When working with polyamides, it is important to examine the water absorption, because it severely affects its mechanical and thermal properties. The density measurements were performed on all composites of polyamide 6/CB/MoS₂. The density values of polyamide 6/CB/MoS₂ composites fall in the range 1.13–1.152 g/cc. Table 2 shows that the density of composites increased linearly with increase in MoS₂ content. MoS₂ being a higher density material, the composites density increases with increase in MoS₂ content. Surface hardness is a measure of resistance to indentation and indicates the degree of compatibility to a certain extent. The surface hardness values of polyamide 6/CB/MoS₂ composites lie in the range 60–70 shore D (Table 2). From the table it is noticed that a significant reduction in surface hardness values to increase in MoS₂ content. The tensile strength and percentage elongation at break increases (Table 2) with increasing in MoS₂ content. The tensile strength of polyamide 6/CB/MoS₂ falls in the range 68–78 MPa. Similarly, percentage elongation at break lies in the range 13.2–22.5% for MoS₂ filled polyamide 6/CB composites. The tensile modulus of the composites increased from 2712 to 2975 MPa. This result indicates that the tensile properties increased significantly with an increase in MoS₂ content in the polyamide 6/CB matrix. Table 2 reveals a reduction in Izod impact

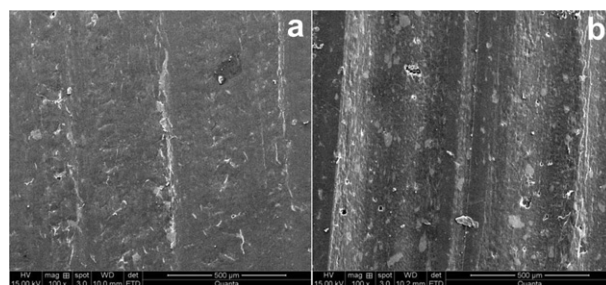


Fig. 7. SEM images of 0.5% MoS₂ filled polyamide 6/CB composite at 9 m s^{−1} sliding velocity, at 50 N load for (a) 1000 m and (b) 2000 m.

Table 2
Physico-mechanical properties of polyamide 6/CB/MoS₂ composites.

Composition (%)			Water uptake (%)	Density (g/cc)	Surface hardness (Shore D)	Tensile strength (MPa)	Elongation at break (%)	Tensile modulus (MPa)	Impact strength (J m ⁻¹)
Polyamide 6	CB	MoS ₂							
98.5	1.0	0.5	2.7	1.130	70	68	13.2	2712	36.5
98.0	1.0	1.0	2.5	1.137	68	73	16.7	2814	34.4
97.0	1.0	2.0	2.4	1.141	63	76	20.8	2944	33.5
96.0	1.0	3.0	2.2	1.152	60	78	22.5	2975	32.9

strength from 36.5 to 32.9 J m⁻¹ with an increase in MoS₂ content. The impact strength of the composites depends upon many factors like toughness of the polymers, degree of miscibility and phase morphology. The nature of the interface region is of extreme importance in determining the toughness of the composites. Here the impact strength decreases gradually with increase in the dosage of MoS₂. This may be due to the addition of MoS₂ reduces the flexibility and energy absorbing capacity of the polyamide 6 matrix by filling the free volume associated with polyamide 6 matrices. That means MoS₂ being a fine powder, it can fill the voids and free volume of polyamide 6 matrices and increase the rigidity. So, with an increase in MoS₂ content, the impact strength decreases.

3.5. Thermal characteristics

The thermal properties of the polyamide 6/CB/MoS₂ composites were investigated by DSC technique to analyze the effect of MoS₂ content on the glass transition temperature (*T_g*) and DSC thermograms are shown in Fig. 8. *T_g* values obtained from the DSC plots are tabulated and shown in Table 3. The *T_g* values increased from 60 to 69 °C with an increase of MoS₂ content from 0.5 to 3.0 wt%, indicating a positive interaction of MoS₂ with amide groups of

polyamide 6. This reveals that segmental mobility of polymer chains is restricted to a certain extent. Similarly the melting temperature of polyamide 6/CB/MoS₂ composites also increased with increase in MoS₂ content as can be seen in Table 3. Polyamide 6 a crystalline material, its cohesive strength and crystallization behavior are associated with the amide groups. The amide groups form strong inter and intra molecular hydrogen bonding. The amide groups and its hydrogen bonding are responsible for the *T_g* and *T_m* associated with polyamide 6. The incorporation of MoS₂ increases the *T_g* and *T_m* of polyamide. The hexagonal form of MoS₂ is characterized by MoS₂ layers in which the Mo atoms have trigonal prismatic co-ordination of six sulfur atoms at the edges. When these MoS₂ layers were inserted into the nylon 6 matrix, the nylon 6 polymer chains near the MoS₂ layers are tightly coiled around the irregularly shaped MoS₂ structure and their mobility is highly restricted. Further the labile protons of the amide (–CO–NH–) groups can have donor-acceptor type interactions. The next layer of polymer chains even though not bound directly to the MoS₂ layers, but strongly bound to the interface layer through hydrogen bonding. So, the incorporation of MoS₂ layers in the nylon 6 matrix restricts the segmental as well as molecular mobility. Hence the *T_g* and *T_m* of nylon 6 increases with increase in MoS₂ dosage. A similar

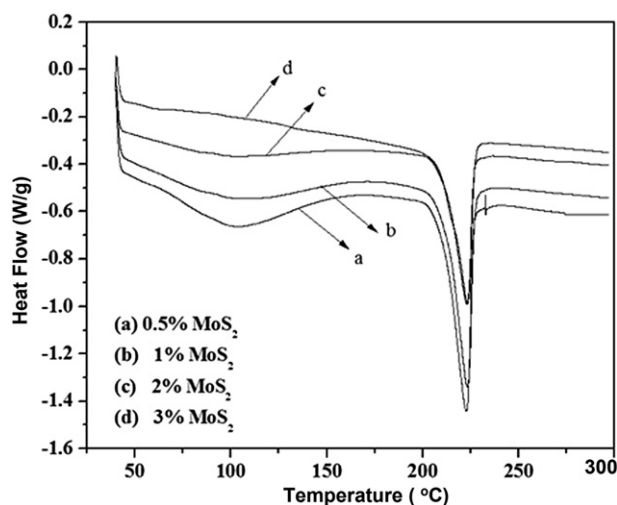


Fig. 8. DSC thermograms of polyamide 6/CB/MoS₂ composites.

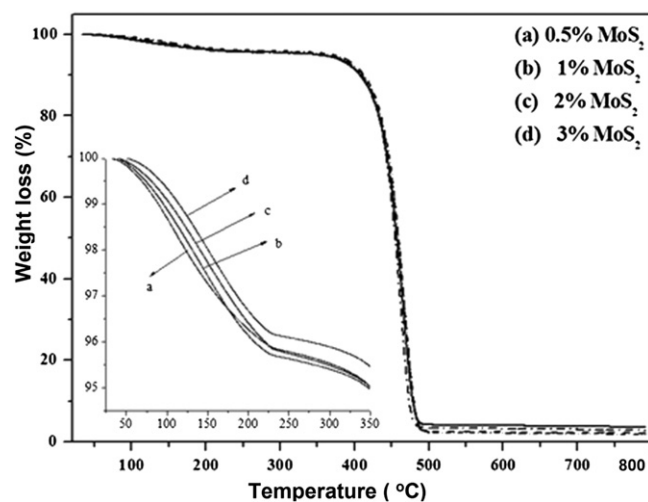


Fig. 9. TGA thermograms of polyamide 6/CB/MoS₂ composites.

Table 3
DSC data obtained for polyamide 6/CB/MoS₂ composites.

Composition (%)			<i>T_g</i> (°C)	<i>T_o</i> (°C)	<i>T_m</i> (°C)	<i>T_c</i> (°C)
Polyamide 6	CB	MoS ₂				
98.5	1.0	0.5	60	205	224	232
98.0	1.0	1.0	62	206	225	234
97.0	1.0	2.0	66	207	227	235
96.0	1.0	3.0	69	208	228	239

Table 4
TGA data obtained for polyamide 6/CB/MoS₂ composites.

Composition (%)			Temperature (°C ± 2)			Ash content (%)
Polyamide 6	CB	MoS ₂	<i>T_o</i>	<i>T_p</i>	<i>T_c</i>	
98.5	1.0	0.5	334	465	508	1.8
98.0	1.0	1.0	343	468	501	2.3
97.0	1.0	2.0	346	470	506	3.4
96.0	1.0	3.0	348	471	512	4.1

Table 5
TGA data obtained for polyamide 6/CB/composites.

Composition (%)			Temperature at different weight loss (± 2 °C)					Oxidation index (OI)
Polyamide 6	CB	MoS ₂	T ₀	T ₁₀	T ₂₀	T ₅₀	T _{max}	
98.5	1.0	0.5	334	408	433	457	479	0.128
98.0	1.0	1.0	343	410	435	459	475	0.162
97.0	1.0	2.0	346	413	436	460	474	0.238
96.0	1.0	3.0	348	415	437	462	473	0.289

trend was observed in DMA analysis that also shows an increase in T_g values with an increase in MoS₂ content.

TG analysis was carried out in order to understand the influence of MoS₂ addition to the thermal stability of the polyamide 6/CB/MoS₂ composites. The TGA thermograms of MoS₂ loaded polyamide 6/CB composites are shown in Fig. 9(a)–(d) along with inset thermograms. The temperature range of thermal degradation was analyzed from the TGA thermograms and is given in Tables 4 and 5. TGA thermograms of the composites indicate single stage thermal degradation process. The decomposition temperature of composites was started at 334 °C and continued up to 512 °C, which corresponds to the weight loss ranging from 95.9 to 98.2%. The obtained percentage ash content is higher in composites as compared to neat polyamide 6 (0.5%). The ash content of polyamide 6/CB/MoS₂ composites increases with increase in MoS₂ content as expected and it lies in the range 1.8–4.2%. From TGA curves it can be clearly observed that the thermal stability markedly improved with increase in MoS₂ content. The improvement in thermal stability is due to the polymer–filler interaction, uniform dispersion and high thermal stability of the MoS₂ layers. The polyamide 6 molecular layers are coiled around the irregularly shaped trigonal prismatic layers of MoS₂ and tightly bound around it. The interaction of amides group with sulfur atoms at the edges of MoS₂ structure also improves the thermal stability of polyamide 6 matrices. Some characteristics TGA data related to the temperature corresponding to weight loss such as T_0 (temperature of onset decomposition), T_{10} (temperature for 10% weight loss), T_{20} (temperature for 20% weight loss), T_{50} (temperature for 50% weight loss) and T_{max} (temperature for maximum weight loss) are the main criteria to indicate the thermal stability of the composites (Table 5). The relative thermal stability of polyamide 6/CB/MoS₂ composites has been evaluated by comparing the decomposition

temperatures at different percentage weight loss (Table 5). The higher the values of T_{10} , T_{20} , T_{50} and T_{max} higher will be the thermal stability of the composites [26]. Fig. 9 and Table 4 data reveal that the initial stage thermal degradation process pattern is almost same for all polyamide 6/CB/MoS₂ composites. The higher the values of oxidation index (OI), higher will be the thermal stability [26–28]. From the table it was observed that the OI values increases with increase in MoS₂ content and it lies in the range 0.128–0.289 (Table 5). This data indicates that the polyamide 6/CB/MoS₂ composites are more thermally stable. Annakutty et al. reported that, the char yield is directly correlated to the potency of flame retardation [29]. From the aforesaid investigation, it can be concluded that the flame resistance of polyamide 6/CB/MoS₂ composites is also enhanced as an increase in MoS₂ content in polyamide 6/CB matrix.

DMA is one of the most appropriate methods to study the viscoelastic behavior and relaxations in polymeric materials. The T_g is a key parameter for most polymers and influences on the use and the processability of the material, possibly more than any other factor. This technique provides very revealing information about these relaxations through the $\tan \delta$ versus temperature curve. $\tan \delta$ is an important parameter characterizing the material's viscoelastic behavior [30]. The same experiment also yields the stiffness (modulus) of the material versus temperature. The obtained storage modulus (Fig. 10) and $\tan \delta$ data along with T_g values of polyamide 6/CB/MoS₂ (Fig. 11) composites are given in Table 6. The storage modulus of polyamide 6 measured at 40 °C was 892 MPa and it increased after incorporating MoS₂ into polyamide 6 matrices. The maximum storage modulus value was noticed to be 1071 MPa for 3 wt% MoS₂ loaded polyamide 6/CB composites, which is about 18% higher than that of pristine polyamide 6. The results obtained in this study are comparable to the literature data [31]. The T_g values of polyamide 6/CB/MoS₂ composites increased from 61 to 65 °C for the increase of MoS₂ content from 0–3.0% and

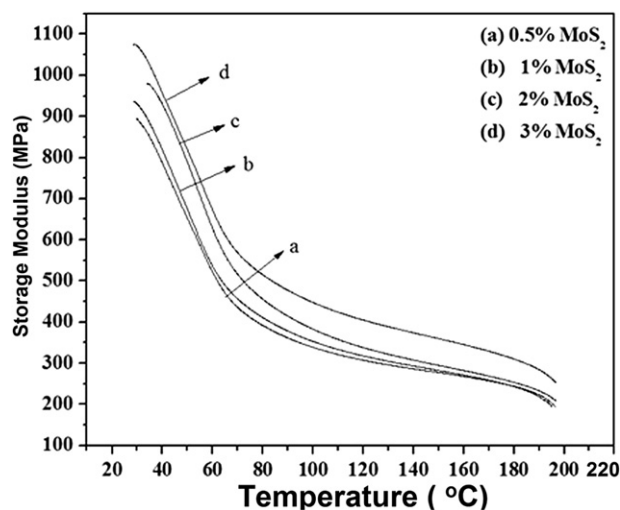


Fig. 10. Plots of storage modulus versus temperature for polyamide 6/CB/MoS₂ composites.

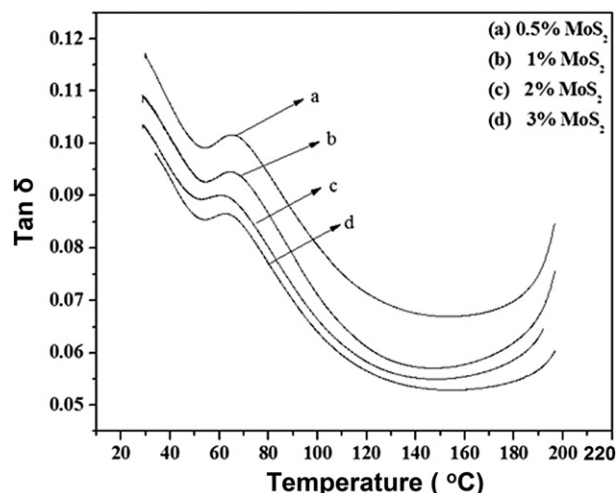


Fig. 11. Plots of tan delta versus temperature for polyamide 6/CB/MoS₂ composites.

Table 6DMA data obtained for polyamide 6/CB/MoS₂ composites.

Composition (%)			Tan δ	T_g (°C)	Storage modulus (MPa)	
Polyamide 6	CB	MoS ₂			Glassy region	Rubbery region
98.5	1.0	0.5	0.102	61.57	892	313
98.0	1.0	1.0	0.095	62.62	930	324
97.0	1.0	2.0	0.090	63.57	977	354
96.0	1.0	3.0	0.087	65.20	1071	418

the T_g of 3 wt% MoS₂ loaded composite shows a positive shift by 4 °C compared to T_g of virgin polyamide 6. While some authors have reported an increase in T_g [32] and others have reported reduction in T_g [33]. The ratio of the loss modulus to the storage modulus, tan δ , increases with increase in MoS₂ content. Polyamide 6 showed a tan δ peak at 61.7 °C attributed the segmental mobility of the resin molecules. The polymer chains nearest to the MoS₂ material are tightly bound and restricted in mobility. The tan δ peak corresponding to T_g values shifted toward higher temperature with the addition of MoS₂ content. This can be explained by the existence of strong interactions between MoS₂ and polyamide 6 matrixes, which limits the movement of the polyamide 6 chain segments. A similar trend was observed in DSC analysis also.

4. Conclusions

The effect of MoS₂ content on tribological and physico-mechanical performances of the polyamide 6/CB/MoS₂ composites have been studied. The MoS₂ content was varied selectively from 0.5 to 3.0 wt% by keeping CB content constant at one percent. Thus, the wear resistance of resultant polyamide 6/CB/MoS₂ composites increased with an increase in MoS₂ content, but reduced with increase in applied load or sliding distances. Similarly the specific wear rate of the polyamide 6/CB/MoS₂ composites decreased with an increase in MoS₂ content, abrading load and sliding distances. The impact strength of the polyamide 6 matrixes decreased from 36.5 to 32.9 J m⁻¹ after incorporation of MoS₂ content. The tensile properties increased gradually with an increase in MoS₂ content. The addition of MoS₂ renders this material somewhat stiffer and dimensionally more stable than pristine polyamide 6, but results in reduction in impact strength. A slight decrease in surface hardness was noticed with an increase in MoS₂ content, this may be attributed to soft MoS₂ content. MoS₂-filled polyamide provides a degree of self-lubrication suited to applications where external lubrication is impractical, contaminating, or difficult to maintain leading to an improvement in wear. This, combined with lower water absorption extends the range of applications that MoS₂-filled polyamide 6. SEM photomicrographs

of polyamide 6/CB/MoS₂ composites at higher load, velocity and distance exhibited wider cracks and deep grooved furrows during the process of surface wear. All of the above research findings suggest that incorporation of two or more filling materials each having a distinct functionality can result in a composite with the potential of enhancing tribological performance via synergistic effect.

References

- [1] Ning Du, C.-Y. Zhao, Q. Chen, G. Wu, R. Liu, Mater. Chem. Phys. 120 (2010) 167–171.
- [2] J. Li, X.H. Cheng, Mater. Chem. Phys. 108 (2008) 67–72.
- [3] S.K. Sinha, Wear Failure of Plastics, in: W.T. Becker, R.J. Shipley (Eds.), ASM Handbook, Failure Analysis and Prevention, vol. 11, ASM International, OH, 2002, pp. 1020–1027.
- [4] J.K. Lancaster, Wear 22 (1972) 412–413.
- [5] Y. Ye, J. Chen, H. Zhou, Wear 266 (2009) 859–864.
- [6] M.H. Cho, S. Bahadur, A.K. Pogorian, Wear 258 (2005) 1825–1835.
- [7] I.M. Hutchings, Tribology, Friction and Wear of Engineering Materials, CRC Press, London, 1992, pp.156–162.
- [8] P.K. Mallick, Fiber Reinforced Composites, Marcel Dekker, Inc., New York, 1993.
- [9] Y. Yamaguchi, Tribology of Plastic Materials, Tribology Series, vol. 16, Elsevier, New York, 1990.
- [10] A M Hager, M Davies, Short-fiber reinforced, high temperature resistant polymers for a wide field of tribological applications, in: K. Friedrich (Ed.) Advances in Composite Tribology, in: R.B. Pipes (Ed.), Composite Materials Series, vol. 8, Elsevier, Amsterdam, The Netherlands, 1993, pp. 107–157.
- [11] S.E. Franklin, Wear 251 (2001) 1591–1598.
- [12] D.-X. Li, X. Deng, J. Wang, J. Yang, X.X. Li, Wear 269 (2010) 262–268.
- [13] Gwidon W. Stachowiak, Wear—Materials Mechanisms and Practice, Wear of Polymer Composites (Friedrich K., Zhang Z. and Klein P.), John Wiley & Sons, Ltd., 2005, pp. 269–271.
- [14] Y. Cohen, D.M. Rein, L. Vaykhansky, Comp. Sci. Technol. 57 (1997) 1149–1154.
- [15] C. Liu, L. Reng, R.D. Amell, J. Tong, Wear 225 (1999) 199–204.
- [16] J. Suwanprateeb, J. Appl. Polym. Sci. 75 (2000) 1503–1513.
- [17] B. Suresha, Siddaramaiah, Kishore, S. Seetharamu, P.S. Kumaran, Wear 267 (2009) 1405–1414.
- [18] E. Basavaraj, Siddaramaiah, J. Macromol. Sci.: Pure Appl. Chem. A47 (2010) 558–563.
- [19] B. Suresha, G. Chandramohan, N.M. Renukappa, Siddaramaiah, J. Appl. Polym. Sci. 103 (2007) 2472–2480.
- [20] J. Chambers, J. Jircny, C. Reese, Fire Mater. 5 (1981) 133–141.
- [21] M. Herrera, G. Matuschek, A. Kettrup, Polym. Degrad. Stab. 78 (2003) 323–331.
- [22] L. Gabriela, E. Avram, G. Paduraru, M. Irimia, N. Hurduc, N. Aelenei, Polym. Degrad. Stab. 82 (2003) 73–79.
- [23] K.S. Annakutty, K. Kishore, Polymer 29 (1988) 756–761.
- [24] T. Jeevananda, Siddaramaiah, Thermochim. Acta 376 (2001) 51–61.
- [25] S. Hamdan, D.M.A. Hashim, Y. Yusop, AJSTD 21 (2003) 69–79.
- [26] C. Zeoer, Elasticity, An Elasticity of Polymer and Metals, The University of Chicago Press, 1956.
- [27] A. Broido, J. Polym. Sci. Part A2 7 (1969) 1761–1773.
- [28] W.O. Winer, Wear 10 (1967) 422–452.
- [29] R. Holinski, J. Gansheimer, Wear 19 (1972) 329–342.
- [30] A.I. Brudnyi, A.F. Karmadonov, Wear 33 (1975) 243–249.
- [31] J.P.G. Farr, Wear 35 (1975) 1–22.
- [32] N. Takahashi, K. Okada, Wear 33 (1975) 153–167.
- [33] N. Takahashi, Wear 124 (1988) 279–289.